

CampusNest – Hostel/PG Finder Platform

Problem Definition

Students who move to Ahmedabad for higher education often face difficulty finding suitable hostel or PG accommodation. Information available through brokers, social media, and personal references is often incomplete, outdated, or difficult to compare. Students have to spend time visiting multiple properties to check rent, location, room type, facilities, and availability. Hostel and PG owners also face difficulties in reaching genuine students and managing property enquiries. A centralized digital platform is needed to connect students, hostel/PG owners, and administrators and make the accommodation search process easier, faster, and more organized.

Features

- User Registration & Login
- Role-Based Access (Student, Hostel/PG Owner, Admin)
- Search Hostel/PG by College Area and Location
- Filters by Rent, Room Type, Gender, and Facilities
- Room Availability Management
- Enquiry / Booking Request Management
- Location and Map Integration
- Admin Dashboard
- Show places nearby hostel/PG (Bus station, hospital, medical store etc)
- Distance between college and their hostel/PG

Technology

Category	Technology / Tools
Frontend	React.js, HTML5, CSS3, JavaScript
Backend	Java 17, Spring Boot (Microservices), Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs
Database	MySQL
Development Tools	Visual Studio Code, MySQL Workbench, Maven, Git & GitHub, Postman

Stakeholders and Outcome

Student: Searches, compares, and enquires about suitable hostels/PGs based on requirements.

Hostel/PG Owner: Adds and manages property listings, room availability, facilities, and student enquiries.

Admin: Manages users, verifies hostel/PG listings, and monitors platform activities.

Expected Outcome

- Make hostel and PG searching easier and faster for students.
- Provide centralized and organized accommodation information.
- Help students compare properties based on rent, location, facilities, and availability.
- Improve communication between students and hostel/PG owners.
- Reduce dependency on brokers and manual property searching.
- Provide verified and transparent hostel/PG listings.
- Enable secure, scalable, and efficient accommodation management through a Spring Boot Microservices architecture.